

To Implement Backslash Escaping Mechanisms for Special Character Processing

CODE:

```
#include <stdio.h>
#include <string.h>

void processEscapes(char *input) {
    int i = 0;

    while (input[i] != '\0') {
        if (input[i] == '\\') {
            i++;

            switch (input[i]) {
                case 'n':
                    printf("\n");
                    break;

                case 't':
                    printf("\t");
                    break;

                case '\\':
                    printf("\\");
                    break;

                case '"':
                    printf("\"");
                    break;

                case '\':
                    printf("'");
                    break;

                case 'r':
                    printf("\r");
                    break;

                case 'b':
                    printf("\b");
                    break;

                default:
                    printf("\\%c", input[i]);
            }
            i++;
        }
    }
}
```

```

    }
    else {
        printf("%c", input[i]);
    }

    i++;
}
}

int main() {
    char input[500];

    printf("Enter a string with escape sequences:\n");
    fgets(input, sizeof(input), stdin);

    printf("\nProcessed Output:\n");
    processEscapes(input);

    return 0;
}

```

Test \n :

```

Hello\nWorld

Processed Output:
Hello
World

```

Test \t :

```

LST@LAPTOP-EK5DCJVE:~# ./escape
Enter a string with escape sequences:
Name:\tTeja

Processed Output:
Name:   Teja

```

Test \\ :

```

LST@LAPTOP-EK5DCJVE:~# ./escape
Enter a string with escape sequences:
C:\\Users\\Teja

Processed Output:
C:\Users\Teja

```

Test quotation marks:

```
LST@LAPTOP-EK5DCJVE:~# ./escape
Enter a string with escape sequences:
He said \"Hello\"

Processed Output:
He said "Hello"
```

Test multiple escapes:

```
He said "Hello"
LST@LAPTOP-EK5DCJVE:~# ./escape
Enter a string with escape sequences:
Hello\nWelcome\tTeja\nPath: C:\\Users\\Teja

Processed Output:
Hello
Welcome Teja
Path: C:\Users\Teja
```